Tech Note T23: Probabilistic error cancellation: paper notes

Summary for key notation of *Probabilistic error cancellation with sparse Pauli-Lindblad models on noisy quantum processors* [Ref. van den Berg et al. (2020)]

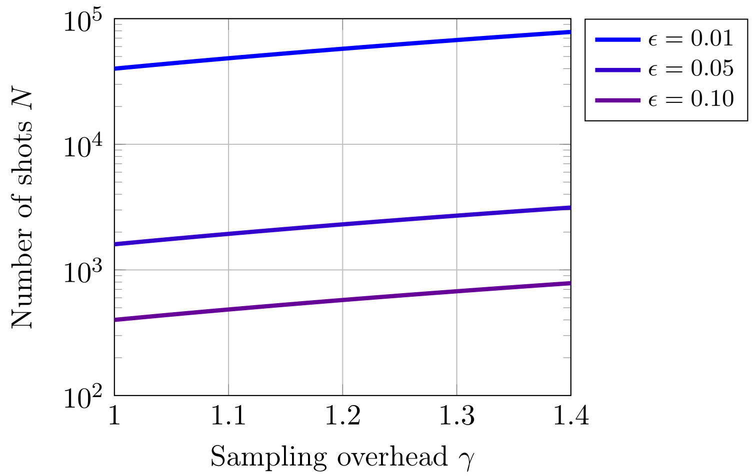
Symbol	Description	Value
	Circuit constants & indices	
n	number of qubits in circuit	$n = 1, 2, 3, \ldots$
l	number of layers in circuit	$l = 1, 2, 3, \ldots$
i	layer index within circuit	$i = 1, \ldots, l$
	Pauli-Lindblad model inputs	
k	model coefficient index, corresponds to a Pauli	$k = 1, 2, 3, \ldots$ or equiv. n-qubit Pauli string
b	fidelity vector indices	$b = 1, 2, \ldots$ or equiv. n-qubit Pauli string
$\mathcal{K}$	set of Pauli fidelity support indices for the sparse model $\mathcal{L}$ of Λ. Small set	$k \in \mathcal{K}$
$\mathcal{B}$	set of benchmark Paulis (Pauli fidelity support indices of Λ). Can be all of them, big set	$b \in \mathcal{B}$
N	number of error-mitigated circuit instances	$N = 1, 2, 3, \ldots$
	Pauli-Lindblad model variables	
γ	sampling overhead	$\gamma \geq 1$, $\gamma = \exp\left(\sum_{k\in\mathcal{K}} 2\lambda_k\right)$
$\gamma_i, \gamma\left(l\right)$	sampling overhead for the i-th layer and a total of l layers	$\gamma\left(l\right) = \Pi_{i=1}^{l}\gamma_i$
λ_k	k-th model coefficient	$\lambda_k \geq 0$
w_k	noise model weight in Λ factoring	$w_k \coloneqq \frac{1}{2}\left(1 + e^{-2\lambda_k}\right)$
f_b	Pauli Λ fidelity of b-th Pauli index	$f_b \coloneqq \frac{1}{2^n}\operatorname{Tr}\left(P_b^{\dagger}\Lambda\left(P_b\right)\right)$
f	vector of Pauli fidelities of Λ	$f = \{f_b\}_{b\in\mathcal{B}}$
$\hat{f}$	fidelity estimates for a set of	
$M(\mathcal{B}, \mathcal{K})$	binary matrix with entries $M_{b,k} = \langle b, k\rangle_{sp}$ where the symplectic product is 0 is the Paulis commute and 1 if they do not	$\log(f) = -2M(\mathcal{B}, \mathcal{K})\lambda$ (element-wise log) (used to fit with $\lambda \geq 0$)
	Quantum operators and superoperators	
P_k	Pauli operator, indexed by k	$P_k \in \{I, X, Y, Z\}^{\otimes n}$
$U, \mathcal{U}, \tilde{\mathcal{U}}_i$	unitary ideal gate; tilde: noisy i-th layer unitary	
Λ, Λ_i	noise channel	$\Lambda\left(\rho\right) = \exp\left[\mathcal{L}\right]\left(\rho\right) = \Pi_{k\in\mathcal{K}}\left(w_k \cdot +(1-w_k)P_k \cdot P_k^{\dagger}\right)\rho$
Λ^{-1}	inverse noise map	$\Lambda^{-1}\left(\rho\right) = \exp\left[-\mathcal{L}\right]\left(\rho\right) = \gamma\Pi_{k\in\mathcal{K}}\left(w_k \cdot -(1-w_k)P_k \cdot P_k^{\dagger}\right)\rho$
$\mathcal{L}\left(\rho\right)$	Lindblad generator	$\mathcal{L}\left(\rho\right) = \sum_{k\in\mathcal{K}}\lambda_k\left(P_k\rho P_k - \rho\right)$
$\left\langle \hat{A}_N \right\rangle$	average error mitigated estimate of $\left\langle \hat{A} \right\rangle$ for some operator $\hat{A}$ using N circuit instances	

23.1 Common questions

How many random circuits to run? For a maximum error bound ϵ between the the ideal and mitigated expectation values $\left|\left\langle \hat{A}\right\rangle_{\text{ideal}} - \left\langle \hat{A}_N\right\rangle\right| \leq \epsilon$ satisfied with a probability $1-\delta$ (assuming a weak enough noise, $C^{l\tau} \approx 1$), we can solve for the number of error-mitigated, random circuit instances N in van den Berg et al. (2020)

$$\begin{aligned} \epsilon &= \gamma \frac{\sqrt{2\log\left(2/\delta\right)}}{\sqrt{N}} \;, \\ \therefore \quad N &= 2\log\left(2/\delta\right)\left(\frac{\gamma}{\epsilon}\right)^2 \;, \\ N &\approx 4\left(\frac{\gamma}{\epsilon}\right)^2 \quad \text{for } \delta \sim 0.01 \;. \end{aligned}$$

For probability $\delta = 2\%$, $\log\left(2/\delta\right) = 2$, a small overhead; for $\delta = 0.01$, it is 2.3, and for $\delta = 0.001$, it is 3.3.



Bibliography

E. van den Berg, Z. K. Minev, and K. Temme, arXiv (2020), ISSN 23318422, 2012.09738, URL http://arxiv.org/abs/2012.09738. Nature Physics **16**, 233 (2020), ISSN 1745-2473, URL https://doi.org/10.1038/s41567-020-0847-3http://www.nature.com/articles/s41567-020-0847-3.

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